

# SPEAKER-GROUP SENSITIVITY OF PAIRED DETECTOR COMPARISONS ON ASVspoof 2021 DF

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## ABSTRACT

A small paired detector advantage can depend strongly on a few observed speaker groups. On fixed ASVspoof 2021 DF scores from eight detectors, we refit equal-error-rate (EER) thresholds under trial and joint speaker-attack resampling and construct simultaneous bands for all 28 EER differences. The number of bands excluding zero falls from 26 to 18. Speaker-only resampling reproduces every primary joint separation decision. Deleting one genuine-only speaker group changes the XLSR-Mamba minus XLSR-Conformer gap from  $-0.389$  to  $-0.129$  percentage points; that group accounts for 77.09% of the squared, mean-centered delete-one deviations for this pair. Concentration varies across the three affected modern comparisons. All sixteen comparisons between the four self-supervised detectors and four organizer baselines survive. A contrasting SpoofCeleb factor pattern limits a universal speaker-only prescription. These are sensitivity results for observed score collections, without population-coverage guarantees.

**Index Terms**— Audio deepfake detection, paired evaluation, speaker influence, EER

## 1. INTRODUCTION

On ASVspoof 2021 DF, XLSR-Mamba achieves 1.884% EER and XLSR-Conformer 2.273%. Deleting VCC2SM3, which contains 315 bona-fide VCC2018 trials and no spoof trials, changes the Mamba-Conformer gap from  $-0.389$  to  $-0.129$  percentage points. Those 315 trials are part of 14,869 bona-fide trials in a benchmark with 533,928 trials overall. The gap remains negative, but its magnitude depends strongly on one observed group.

This deletion belongs to a complete family analysis, with all speakers retained in the principal comparison. Across eight detectors, we ask which paired EER differences survive changing the resampling unit, and how concentrated the delete-one speaker influence is in the affected modern comparisons. ASVspoof 2021 DF has 93 speakers and 110 attacks; a large trial count need not provide broad group support for a close comparison. Shared group variation may cancel between detectors, so dependence of individual scores alone cannot answer this paired question.

Our contribution is an empirical concentration result on fixed scores: speaker-only resampling reproduces the joint

decisions, and delete-one influence is concentrated to differing degrees across the three affected self-supervised (SSL) comparisons. We print every primary trial and joint band, then test comparison-family and source-composition objections. A limited SpoofCeleb counterexample retains the reason to examine both factors. The reporting consequence is to disclose influential-group sensitivity when claiming a small paired EER advantage.

## 2. RELATED WORK

**Benchmark reporting.** ASVspoof 2017 reported EER intervals [1]; ASVspoof 2021 Fig. 4(c) used the Bengio-Mariéthoz test [2] with Holm correction [3]. Its shared-trial variant accounts for two systems using the same trials, not repeated speakers and attacks. We found no uncertainty analysis in the later journal overview [4] or ASVspoof 5 [5]; a recent SSL benchmark reports point EERs [6]. Other work reports trial-bootstrap ranks [7], EER spreads [8], corrected pairwise tests [9], or fixed-threshold intervals [10]. Training-seed variation also moves rankings [11]; our scores hold training fixed.

**Dependent paired comparison.** User-aware biometric and speech resampling [12, 13], correlation-corrected paired tests [14], and synchronized resampling for correlated classifiers [15] precede this study. Our speaker and spoof-attack factors are crossed [16, 17, 18], distinct from the nested-observation designs studied in [19]. We use these established tools to measure concentrated group influence on paired EER comparisons, with every threshold refitted.

**Evaluation design.** Aggregation weights [20], channel presentation [21], representations [22], controlled generator/speaker perturbations [23] and corpus overlap [24] affect deepfake evaluation. Identity perturbations [25], speaker-adversarial training [26] and speaker/synthesizer familiarity [27] address related sensitivities. Here the scores are fixed and evaluation weights vary. Simultaneous rank intervals are established [28]. Because selecting only neighbors in the observed point-EER ordering uses the outcomes, we include every detector pair.

### 3. METHOD

#### 3.1. EER contrast and separation event

Two deterministic systems on a fixed trial list have no sampling uncertainty until a perturbation law is declared. Scores are oriented so larger means bona fide and sorted in ascending order. At each ordered position, the false-reject rate (FRR) is the fraction of bona-fide weight at or below it and the false-accept rate (FAR) the fraction of spoof weight above it; we select the first position minimizing their absolute difference and take their mean as EER, without interpolation. For ordered pair  $(A, B)$ ,  $\Delta\text{EER}_{A,B} = 100(\text{EER}_A - \text{EER}_B)$  percentage points (negative favors  $A$ ); every replicate refits both thresholds. A pair is *separated under procedure  $h$*  when its simultaneous band excludes zero; write this indicator as  $I_{p,h}$ .

#### 3.2. Perturbation laws and scope

The trial bootstrap independently resamples trials within class; for SpoofCeleb only, the archived plan samples the 91,130 trial indices from the pooled list with replacement, so class counts vary across replicates. In the speaker $\times$ attack product-weight (PW) bootstrap [18, 29], we independently draw speakers and spoof attacks uniformly with replacement, drawing as many of each as observed; a spoof trial receives the product of its speaker and attack multiplicities, and a bona-fide trial receives only its speaker multiplicity. Speaker-only fixes attack multiplicities at one; attack-only fixes speaker and bona-fide trial weights at one. Each of these four procedures uses 5,000 replicates. Each replicate’s weights are shared across systems; procedures have separate random streams.

**Interpretation.** These bands describe sensitivity of fixed scores to the stated weights, not confidence about future speakers or attacks. Four VCC spoof strata contain only 4, 4, 4 and 6 speakers. Coverage results for crossed-array means do not establish coverage for refitted EER differences [29]. The one-factor, Monte Carlo, mean/sign, pairing, tie and group-deletion diagnostics were designed after the primary results and are exploratory. Matching separation decisions does not decompose variance or identify a causal speaker effect. A band containing zero establishes neither equality nor a deployment ranking.

#### 3.3. All-pair simultaneous bands

For procedure  $h$ , let  $s_{p,h} = \text{sd}_b(\Delta_{p,h}^{(b)})$ . We set  $q_{.95,h}$  to the empirical 95th percentile of  $\max_{p=1}^{28} |\Delta_{p,h}^{(b)} - \bar{\Delta}_{p,h}|/s_{p,h}$  and report  $\hat{\Delta}_p \pm q_{.95,h}s_{p,h}$  [30]. The quantile centers deviations on the bootstrap mean, whereas the bands center on the original estimate; band inclusion is not an empirical reversal probability. The maximum includes all 28 primary contrasts, including when reporting within-cohort counts; SpoofCeleb

uses its own complete six-pair family. No population error-control guarantee is asserted.

#### 3.4. Delete-one concentration

For each of the 93 speakers, we remove all its trials, renormalize within class and refit both thresholds. For pair  $p$ , define  $d_{p,g} = \hat{\Delta}_{p,-g} - \bar{\Delta}_{p,-\cdot}$ , where the bar averages the 93 delete-one estimates. The influence share is  $d_{p,g}^2 / \sum_{h=1}^{93} d_{p,h}^2$ ; top-five shares sum the five largest terms of this same denominator. These are shares of squared deletion deviations, not fractions of the original gap or population effective sample sizes.

## 4. RESULTS

**Setup.** The primary set contains XLSR-Mamba [31], XLSR+SLS (SLS below) [32], XLSR-Conformer [33] and SSL-AASIST [34], plus RawNet2, LFCC-LCNN, LFCC-GMM and CQCC-GMM. All 533,928 evaluation-phase trials have every score: 14,869 bona-fide and 519,059 spoof trials, with weights normalized within class. The matched trial/PW comparison uses 5,000 replicates and seed 2026081604; NumPy SeedSequence children are trial, joint, speaker-only and attack-only in that order.

**Complete paired evidence.** The matched all-28-pair bootstrap refits the same weighted EER in every replicate. Table 1 prints all point differences and both simultaneous bands. Trial and joint bands separate 26/28 and 18/28 pairs; within SSL, the counts fall from 5/6 to 2/6. All primary within-SSL losses involve Conformer. All sixteen SSL-versus-baseline comparisons remain separated under every primary arm (Table 3). Restricting the maximum to the six SSL pairs also gives 5/6 trial and 2/6 joint separations; the losses do not require the organizer comparisons in the family.

**Concentration across affected SSL pairs.** Table 2 reports the largest speaker contributor for each SSL comparison losing separation. The two comparisons of Mamba and SLS with Conformer share VCC2SM3 as their largest contributor, whereas Conformer-SSL-AASIST has a different and less dominant contributor. All speakers remain in the principal analysis. Deletion changes class normalization and refits thresholds; it measures influence without isolating a causal speaker effect or an exclusive tail mechanism.

**Factor pattern and pairing.** Speaker-only reproduces every primary joint separation decision; attack-only reproduces four or five of the eight joint losses across Monte Carlo repeats. Matching indicators does not make attacks inert: Table 4 shows different SDs and opposite-sign counts under speaker-only and joint resampling. Under the joint law, Mamba and Conformer EERs correlate 0.777; pairing removes about 75% of the sum of their marginal variances, yet the paired SD rises from 0.06149 under trials to 0.31997

**Table 1.** Complete primary 21DF sensitivity bands in percentage points, using the full 28-pair maximum. Pair labels mean first minus second; negative favors the first. A star marks trial separation lost under joint resampling. Abbreviations and point EERs (%) are defined above the bands.

|   |                |       |   |           |        |
|---|----------------|-------|---|-----------|--------|
| M | XLSR-Mamba     | 1.884 | R | RawNet2   | 22.383 |
| S | XLS-R+SLS      | 1.916 | L | LFCC-LCNN | 23.477 |
| C | XLSR-Conformer | 2.273 | F | LFCC-GMM  | 25.247 |
| A | SSL-AASIST     | 2.852 | Q | CQCC-GMM  | 25.563 |

  

| Pair                           | Gap     | Trial band         | Joint band         |
|--------------------------------|---------|--------------------|--------------------|
| Within SSL                     |         |                    |                    |
| M-S                            | -0.032  | [-0.144, 0.080]    | [-0.725, 0.661]    |
| M-C*                           | -0.389  | [-0.570, -0.208]   | [-1.348, 0.570]    |
| M-A                            | -0.968  | [-1.175, -0.760]   | [-1.828, -0.107]   |
| S-C*                           | -0.357  | [-0.538, -0.177]   | [-1.493, 0.779]    |
| S-A                            | -0.935  | [-1.141, -0.730]   | [-1.784, -0.087]   |
| C-A*                           | -0.578  | [-0.835, -0.322]   | [-1.637, 0.480]    |
| Within organizer baselines     |         |                    |                    |
| R-L*                           | -1.094  | [-1.722, -0.466]   | [-5.528, 3.341]    |
| R-F*                           | -2.864  | [-3.654, -2.074]   | [-9.949, 4.221]    |
| R-Q*                           | -3.180  | [-4.003, -2.356]   | [-9.284, 2.925]    |
| L-F*                           | -1.770  | [-2.426, -1.114]   | [-6.490, 2.950]    |
| L-Q*                           | -2.086  | [-2.765, -1.407]   | [-6.173, 2.001]    |
| F-Q                            | -0.316  | [-0.929, 0.297]    | [-3.370, 2.738]    |
| SSL versus organizer baselines |         |                    |                    |
| M-R                            | -20.499 | [-21.070, -19.929] | [-26.950, -14.049] |
| M-L                            | -21.593 | [-22.123, -21.063] | [-27.445, -15.742] |
| M-F                            | -23.363 | [-23.981, -22.746] | [-29.462, -17.264] |
| M-Q                            | -23.679 | [-24.364, -22.994] | [-29.247, -18.112] |
| S-R                            | -20.467 | [-21.039, -19.896] | [-26.958, -13.976] |
| S-L                            | -21.561 | [-22.090, -21.032] | [-27.440, -15.682] |
| S-F                            | -23.331 | [-23.948, -22.714] | [-29.469, -17.194] |
| S-Q                            | -23.647 | [-24.330, -22.964] | [-29.216, -18.078] |
| C-R                            | -20.110 | [-20.673, -19.547] | [-26.307, -13.914] |
| C-L                            | -21.204 | [-21.726, -20.682] | [-26.757, -15.651] |
| C-F                            | -22.974 | [-23.587, -22.361] | [-28.899, -17.049] |
| C-Q                            | -23.290 | [-23.964, -22.616] | [-28.709, -17.871] |
| A-R                            | -19.532 | [-20.118, -18.945] | [-25.739, -13.325] |
| A-L                            | -20.626 | [-21.171, -20.080] | [-26.215, -15.036] |
| A-F                            | -22.396 | [-23.028, -21.763] | [-28.399, -16.392] |
| A-Q                            | -22.712 | [-23.406, -22.017] | [-28.123, -17.301] |

**Table 2.** Speaker deletion influence for the three affected SSL pairs (abbreviations as in Table 1). Gaps are in points; shares use the squared-deviation denominator in Method.

| Pair | Largest group | Full → deletion | Largest / top five |
|------|---------------|-----------------|--------------------|
| M-C  | VCC2SM3       | -0.389 → -0.129 | 77.09% / 90.03%    |
| S-C  | VCC2SM3       | -0.357 → -0.131 | 55.85% / 78.47%    |
| C-A  | VCC2SF2       | -0.578 → -0.441 | 23.85% / 75.73%    |

points under joint resampling. The trial and joint critical values are 2.940 and 2.997, so SD inflation drives the band widening. The joint mean displacement is only 0.032 SD, so

**Table 3.** Primary zero-exclusion counts under the full 28-pair correction; every arm retains 16/16 SSL-versus-baseline separations. Attack ranges span the original and five fresh 5,000-draw runs; other entries are recorded-run counts. One-factor analyses are exploratory.

| Roster / subset | Trial | Speaker | Attack   | Joint |
|-----------------|-------|---------|----------|-------|
| 21DF, all       | 26/28 | 18/28   | 21-22/28 | 18/28 |
| 21DF, SSL       | 5/6   | 2/6     | 3/6      | 2/6   |
| 21DF, organizer | 5/6   | 0/6     | 2-3/6    | 0/6   |

**Table 4.** Mamba minus Conformer: mean shift from the point gap, pair SD (points), opposite-sign draws out of 5,000, and zero exclusion under the full 28-pair band.

| Law     | Mean shift | SD       | Opposite | Excludes? |
|---------|------------|----------|----------|-----------|
| Trial   | -0.002492  | 0.061488 | 0        | Yes       |
| Speaker | -0.003175  | 0.270826 | 210      | No        |
| Attack  | -0.010235  | 0.147333 | 37       | No        |
| Joint   | -0.010135  | 0.319971 | 463      | No        |

band centering does not explain inclusion of zero; the trial band excludes zero with 0/5,000 opposite-sign draws, the joint band includes it with 463/5,000. A finite zero count does not establish zero reversal probability.

**Composition-preserving control.** A post-result PW control resamples speakers and attacks within each bona-fide source or spoof stratum, rejects draws with no class support in any required stratum, and rescales the three bona-fide source masses and five spoof-stratum masses to their observed values. This control uses seed 20260824. The bona-fide sources are ASVspoof, VCC2018 and VCC2020; the spoof strata split VCC2018 into HUB/SPO and VCC2020 into Task 1/Task 2. VCC2018 required 1,007 attempts, with seven rejected for lost spoof support; no other stratum required rejections. Across 1,000 retained draws it separates 18/28 pairs and 0/6 organizer pairs. For Mamba-Conformer, the control band is [-1.280, 0.501], beside the primary joint band [-1.348, 0.570]. Thus movement of these eight masses is not required for the loss of separation; this does not remove influence from groups within a source or stratum.

**Numerical checks.** Across five fresh attack-only runs, separation counts range from 21 to 22 overall and from two to three among organizer baselines; within-SSL counts remain three. Five fresh 5,000-draw runs for each of the trial, speaker-only and joint arms also preserve every recorded separation indicator. A distinct-score-boundary tie check over the four primary arms and five fresh attack streams changes no indicator; the largest pair-band endpoint change is 0.000233 points.

**Counterexample on single-source SpoofCeleb.** On all 91,130 SpoofCeleb evaluation trials [35], pooled trial resampling separates 6/6 pairs and joint resampling 3/6;

**Table 5.** SpoofCeleb: all six pairs under their own six-pair maximum. T/S/A/J denote trial/speaker/attack/joint; Y means the band excludes zero, N means it includes zero. SSL-A denotes SSL-AASIST. Below: point gaps and trial/joint bands (points) for the three joint losses.

| Pair         | T   | S   | A   | J   |
|--------------|-----|-----|-----|-----|
| AASIST-SLS   | Y   | Y   | Y   | Y   |
| AASIST-SSL-A | Y   | Y   | Y   | Y   |
| AASIST-Mamba | Y   | Y   | Y   | Y   |
| SLS-Mamba    | Y   | Y   | N   | N   |
| SSL-A-SLS    | Y   | Y   | N   | N   |
| SSL-A-Mamba  | Y   | N   | N   | N   |
| Separated    | 6/6 | 5/6 | 3/6 | 3/6 |

  

| Pair        | Gap    | Trial            | Joint             |
|-------------|--------|------------------|-------------------|
| SLS-Mamba   | -3.063 | [-3.771, -2.355] | [-16.062, 9.936]  |
| SSL-A-SLS   | 2.207  | [1.592, 2.821]   | [-7.392, 11.806]  |
| SSL-A-Mamba | -0.856 | [-1.446, -0.266] | [-14.805, 13.093] |

class-stratified trial resampling also gives 6/6. EERs are 57.93%, 24.51%, 26.72% and 27.58% for AASIST, SLS, SSL-AASIST and Mamba. The roster has 40 speakers and nine attacks. Trial/joint arms use seed 20260829; the post-result one-factor check uses seed 20260830. Every base item has one bona-fide and nine spoof renditions under one source label, so between-source mass cannot change. Under the original six-pair correction, all three comparisons among SLS, SSL-AASIST and Mamba lose separation; the three AASIST contrasts retain it. Table 5 prints the four-arm decisions and bands for the joint losses. Here attack-only reproduces the joint indicator vector, while speaker-only separates 5/6 pairs. This is a control on a weakly transferred, off-domain roster, not evidence that attack effects generally dominate speaker effects on another corpus. Different rosters and poor transfer prevent a causal attribution of this contrast to the corpus; the mirrored factor pattern nevertheless counters a universal speaker-only prescription.

## 5. DISCUSSION AND CONCLUSION

The contribution is measured concentration across paired comparisons on this fixed benchmark, not a new confidence procedure. A diffuse delete-one profile across the affected SSL pairs would contradict the concentration claim; Table 2 exposes both the strong flagship case and the weaker third pair. A changed speaker-only decision would contradict the reported primary factor agreement. Neither finding establishes why those groups matter, and SpoofCeleb shows why the factor pattern should not become a universal prescription.

For a small paired EER advantage on this benchmark, report the point difference, trial and group bands for the same family, and influential-group deletion sensitivity. If only the

trial band excludes zero, describe the advantage as sensitive to evaluation composition and seek broader group support before claiming robust superiority. Publish the group memberships and weighting rules needed to reproduce the check. The sixteen SSL-versus-baseline contrasts survive all four primary resampling laws, while close within-cohort comparisons are sensitive. These results support reporting sensitivity to observed-group weights, without establishing population confidence or a universal factor pattern across corpora.

**Artifact.** Code, score provenance, complete numerical results and additional reproducibility checks are available in the public artifact. Artifact version

260e1e15e1ac6f79d4a7b083f5a51716dc8d7516  
at  
github.com/rvirgilli/asvspoof2021-df-clustered-uncertainty.

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## 7. COMPLIANCE WITH ETHICAL STANDARDS

This study analyzed previously collected ASVspoof 2021 and SpoofCeleb recordings and detector scores. It involved no new recording or interaction with human participants; no ethical approval was required. The authors declare no competing interests.

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